

Experiment:4

Construct a scheduling program with C that selects the waiting process with the smallest execution time to execute next.

Code:-

```
#include <stdio.h>

int main()
{
    int bt[20], p[20], wt[20], tat[20];
    int i, j, n, total = 0, pos, temp;
    float avg_wt, avg_tat;
    printf("Enter number of processes: ");
    scanf("%d", &n);
    printf("Enter Burst Time:\n");
    for (i = 0; i < n; i++)
    {
        printf("P%d: ", i + 1);
        scanf("%d", &bt[i]);
        p[i] = i + 1;
    }
    for (i = 0; i < n; i++)
    {
        pos = i;
        for (j = i + 1; j < n; j++)
        {
            if (bt[j] < bt[pos])
                pos = j;
        }
    }
}
```

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}
temp = bt[i];
    bt[i] = bt[pos];
    bt[pos] = temp;
temp = p[i];
    p[i] = p[pos];
    p[pos] = temp;
}
wt[0] = 0;
for (i = 1; i < n; i++)
{
    wt[i] = 0;
for (j = 0; j < i; j++)
    wt[i] += bt[j];

    total += wt[i];
}
avg_wt = (float)total / n;
total = 0;
printf("\nProcess\tBurst Time\tWaiting Time\tTurnaround Time\n");

for (i = 0; i < n; i++)
{
    tat[i] = bt[i] + wt[i];
    total += tat[i];

    printf("P%d\t%d\t\t%d\t\t%d\n",

```

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        p[i], bt[i], wt[i], tat[i]);
    }

    avg_tat = (float)total / n;

    printf("\nAverage Waiting Time = %.2f", avg_wt);
    printf("\nAverage Turnaround Time = %.2f\n", avg_tat);

    return 0;
}

```

Output:

```

Enter number of processes: 3
Enter Burst Time:
P1: 8
P2: 5
P3: 3

P      BT      WT      TAT
P3     3       0       3
P2     5       3       8
P1     8       8      16

Average Waiting Time = 3.67
Average Turnaround Time = 9.00

...Program finished with exit code 0
Press ENTER to exit console.

```